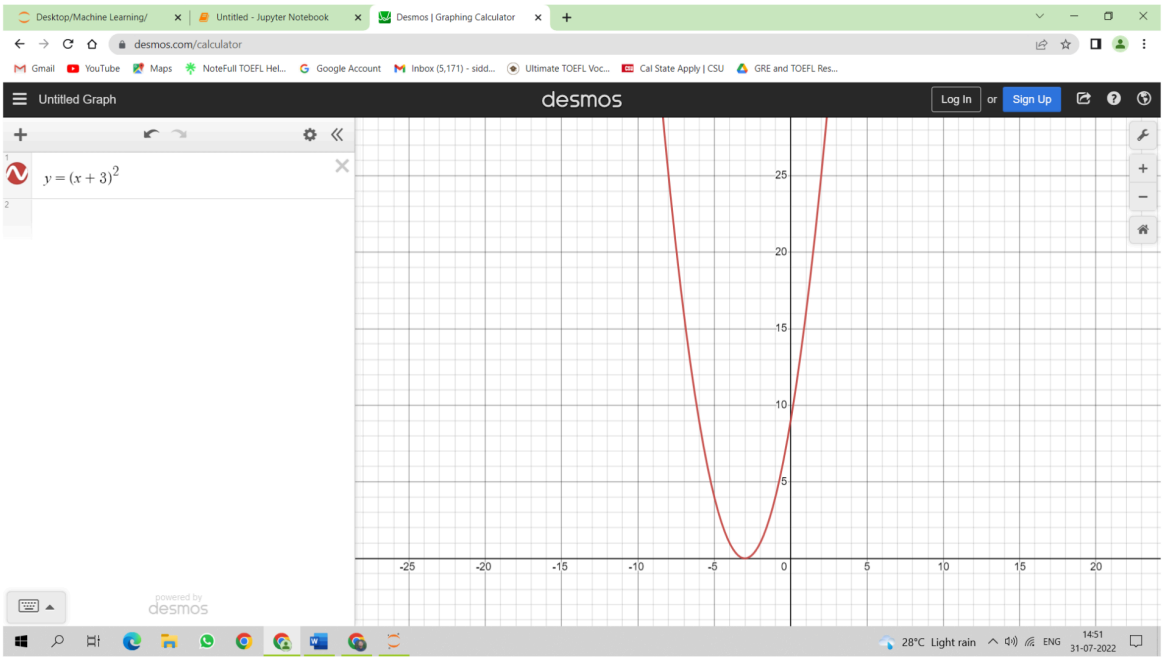
Implement Gradient Descent Algorithm to find the local minima of a function.

For example, find the local minima of the function $y=(x+3)^2$ starting from the point x=2.



We know the answer just by looking at the graph. $y = (x+3)^2$ reaches it's minimum value when x = -3 (i.e when x=-3, y=0). Hence x=-3 is the local and global minima of the function.Below is the implementation in python

In [1]:
```
current_x = 2 # The algorithm starts at x=3
rate = 0.01 # Learning rate
precision = 0.000001 #This tells us when to stop the algorithm
previous_step_size = 1
max_iters = 10000 # maximum number of iterations
iters = 0 #iteration counter
df = lambda x: 2*(x+3) #Gradient of our function
```

In [6]:
```
while previous_step_size > precision and iters < max_iters: #When Previous Step SIze will be less than Precision then we will reach the global maxima
    previous_x = current_x #Store current x value in prev_x
    current_x = current_x - rate * df(previous_x) #Grad descent
    previous_step_size = abs(current_x - previous_x) #Change in x
    iters = iters+1 #iteration count
    print("Iteration",iters,"\nX value is",current_x) #Print iterations

print("The local minimum occurs at", current_x)
```

```
Iteration 1
X value is 1.8019999999999998
Iteration 2
X value is 1.70596
Iteration 3
X value is 1.6118408
Iteration 4
X value is 1.519603984
Iteration 5
X value is 1.42921190432
Iteration 6
X value is 1.3406276662336
Iteration 7
X value is 1.2538151129088928
Iteration 8
X value is 1.1687388106507495
Iteration 9
X value is 1.0853640344377344
Iteration 10
X value is 1.0036567537489798
Iteration 11
X value is 0.9235836186740002
Iteration 12
X value is 0.8451119463005202
Iteration 13
X value is 0.7682097073745098
Iteration 14
X value is 0.6928455132270196
Iteration 15
X value is 0.6189886029624792
Iteration 16
X value is 0.5466088309032295
Iteration 17
X value is 0.4756766542851649
Iteration 18
X value is 0.40616312119946163
Iteration 19
X value is 0.3380398587754724
Iteration 20
X value is 0.27127906159996296
Iteration 21
X value is 0.2058534803679637
Iteration 22
X value is 0.14173641076060442
Iteration 23
X value is 0.07890168254539233
Iteration 24
X value is 0.017323648894484484
Iteration 25
X value is -0.04302282408340521
Iteration 26
X value is -0.10216236760173711
Iteration 27
X value is -0.16011912024970237
Iteration 28
X value is -0.21691673784470833
Iteration 29
X value is -0.27257840308781417
Iteration 30
X value is -0.3271268350260579
Iteration 31
X value is -0.38058429832553675
Iteration 32
X value is -0.432972612359026
Iteration 33
X value is -0.4843131601118455
Iteration 34
X value is -0.5346268969096086
Iteration 35
X value is -0.5839343589714164
Iteration 36
X value is -0.632255671791988
Iteration 37
X value is -0.6796105583561483
Iteration 38
X value is -0.7260183471890252
Iteration 39
X value is -0.7714979802452447
Iteration 40
X value is -0.8160680206403399
Iteration 41
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Iteration 42
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Iteration 43
X value is -0.9445006924825228
Iteration 44
X value is -0.9856106786328723
Iteration 45
X value is -1.025898465060215
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X value is -1.0653804957590107
Iteration 47
X value is -1.1040728858438305
Iteration 48
X value is -1.1419914281269539
Iteration 49
X value is -1.1791515995644148
Iteration 50
X value is -1.2155685675731265
Iteration 51
X value is -1.2512571962216639
Iteration 52
X value is -1.2862320522972306
Iteration 53
X value is -1.320507411251286
Iteration 54
X value is -1.3540972630262602
Iteration 55
X value is -1.387015317765735
Iteration 56
X value is -1.4192750114104202
Iteration 57
X value is -1.4508895111822118
Iteration 58
X value is -1.4818717209585675
Iteration 59
X value is -2.3369822285262357
Iteration 100
X value is -2.3501641761157104
Iteration 101
X value is -2.3631608925933962
Iteration 102
X value is -2.375897674741528
Iteration 103
X value is -2.3883797212466975
Iteration 104
X value is -2.4006121268217635
Iteration 105
X value is -2.4125998842853282
Iteration 106
X value is -2.4243478865996217
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X value is -2.4358609288676294
Iteration 108
X value is -2.447143710290277
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X value is -2.458200836084471
Iteration 110
X value is -2.4690368193627816
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X value is -2.479656082975526
Iteration 112
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Iteration 113
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Iteration 114
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Iteration 116
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Iteration 123
X value is -2.591677426217222
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X value is -2.6078470001390204
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X value is -2.61569006013624
Iteration 127
X value is -2.623376258933515
Iteration 128
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Iteration 129
X value is -2.638290559079748
Iteration 130
X value is -2.6455247478981527
Iteration 131
X value is -2.6526142529401895
Iteration 132
X value is -2.6595619678813858
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X value is -2.666370728523758
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X value is -2.6730433139532828
Iteration 135
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Iteration 136
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Iteration 139
X value is -2.7044570518295643
Iteration 140
X value is -2.7103679107929732
Iteration 141
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Iteration 148
X value is -2.7535917283495523
Iteration 149
X value is -2.7585198937825615
Iteration 150
X value is -2.76334949590691
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Iteration 535
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